

Dimensional Reduction in Supergravity and Duality Symmetries

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Overview

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- 2 $D + E$ gravity description
- 3 $O(E, E)$ symmetry
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Other force

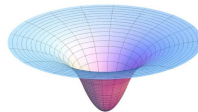
quantum field-theoretic language

Other force

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General relativity

Gravitational interaction due to
space time deformation

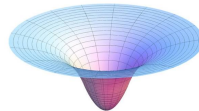


Other force

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The classical quantification method do not work on gravity. We
need to modify the gravity description.

supersymmetry

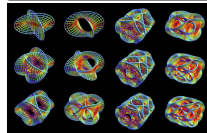
symmetry mixing the degrees of freedom of fermions and bosons.

Used in :

Supergravity



String Theory



supersymmetry

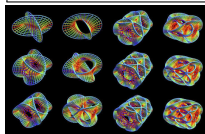
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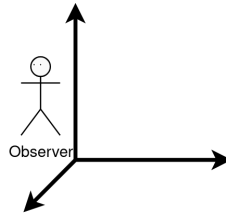
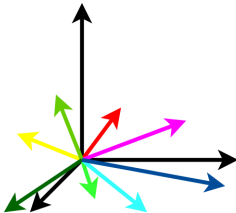


String Theory



!!Both theories describe the existence of extra dimensions!!

We need a way to describe how 4 dimensions observer, see a $4 + E$ dimensions universe.



4 dimension gravity

- We describe GR as a deformation of space-time describe in a metric:

$$g_{\mu\nu} = \begin{pmatrix} g_{00} & g_{01} & g_{02} & g_{03} \\ g_{01} & g_{11} & g_{12} & g_{13} \\ g_{02} & g_{12} & g_{22} & g_{23} \\ g_{03} & g_{13} & g_{23} & g_{33} \end{pmatrix}$$

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- Einstein eqation: $R_{\mu\nu} \frac{1}{2} g_{\mu\nu} R = 0$.

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This give us the hilbert-Einstein action :

$$S_g = \int dx^4 \sqrt{-g} R$$

electromagnetic 4 dimensions description

Maxwell equations

- $\vec{\nabla} \cdot \vec{E} = \mu_0 c^2 \rho,$
- $\vec{\nabla} \times \vec{B} = \mu_0 \vec{j} + \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t},$
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Covariant formulation

- Four-current: $J^\mu = (c\rho, \vec{j}),$
- Four-potential: $A^\mu = (V/c, \vec{A}),$
- field strength: $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$

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This give us the electromagnetic action or Maxwell action in a vaccum:

$$S_F = \int dx^4 \frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

$D + E$ dimensions metric

If we add new dimension the action have to be modified:

$$S_{\hat{g}} = \int dx^D \int \frac{dy^E}{\rho(E)} \sqrt{-\hat{g}} \hat{R}$$

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and the action will have to be expended, for example in 5 dimension we have:

$$\hat{g}_{\mu\nu} = \begin{pmatrix} g_{00} & g_{01} & g_{02} & g_{03} & g_{04} \\ g_{01} & g_{11} & g_{12} & g_{13} & g_{14} \\ g_{02} & g_{12} & g_{22} & g_{23} & g_{24} \\ g_{03} & g_{13} & g_{23} & g_{33} & g_{34} \\ g_{04} & g_{14} & g_{24} & g_{34} & g_{44} \end{pmatrix}$$

Reduction

We are going to follow the reduction method of the Scherk and Schwarz, “How to get masses from extra dimensions”. Firstly let's do an ansatz of the metric we are going to use

$$\hat{g}_{\hat{\mu}\hat{\nu}} = \begin{pmatrix} g_{\mu\nu} - A_{\mu}^{\gamma} A_{\nu\lambda}^{(1)} & -A_{\mu\beta}^{(1)} \\ A_{\nu\alpha}^{(1)} & -h_{\alpha\beta} \end{pmatrix}$$

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And once finished that gave us:

$$S_{\hat{g}} = \int dx^D e^{-\hat{\phi}} V \delta \left(\frac{-1}{4} g^{\rho\lambda} \partial_{\lambda} h_{\alpha\beta} \partial_{\rho} h^{\alpha\beta} + \frac{1}{4} F^{\mu\nu\alpha} F_{\mu\nu}^{\beta} h_{\alpha\beta} + \right. \\ \left. + R + \frac{1}{4} \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \log(h) + \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \hat{\phi} \right).$$

The NS-NS action

In string theory we do not use the Einstein Hilbert action but a variant of it the NS-NS action:

$$S = \int d x^D \int \frac{d y^E}{\rho(E)} \sqrt{-\hat{g}} e^{-\hat{\phi}} \left(\hat{R} + \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\mu}} \hat{\phi} - \frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \right),$$

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$\hat{\phi}$ is the dilaton, and $\hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}}$ is the strength field of the two form $\hat{B}_{\mu\nu}$. We define the low components of $\hat{B}_{\mu\nu}$ with:

$$T^\mu = V_r^\mu \hat{V}_{\hat{\mu}}^r \hat{T}^{\hat{\mu}}, \quad (1a)$$

$$T^\alpha = V_a^\alpha \hat{V}_{\hat{\mu}}^a \hat{T}^{\hat{\mu}}, \quad (1b)$$

$$T_\mu = V_\mu^r \hat{V}_r^{\hat{\mu}} \hat{T}_{\hat{\mu}}, \quad (1c)$$

$$T_\alpha = V_\alpha^a \hat{V}_a^{\hat{\mu}} \hat{T}_{\hat{\mu}}. \quad (1d)$$

An action reformed

We can now write the Reduced action :

$$S = \int dx^D V e^{-\phi} \left(R + \partial_\mu \phi \partial^\mu \phi + \frac{1}{8} \text{tr}(\partial_\mu M^{-1} \partial^\mu M) - \frac{1}{4} \mathcal{F}_{\mu\nu}^i (M^{-1})_{ij} \mathcal{F}^{\mu\nu j} - \frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \right).$$

with M being:

$$M = \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix}.$$

$O(E, E)$ invariance

We define the action of the $O(E, E)$ group as being,

$$A^i \rightarrow \Omega_j^i A^j,$$

$$M_{ij} \rightarrow \Omega_i^l M_{lk} \Omega_j^k,$$

where $\Omega \in O(E, E)$.

The action is invariant under this transformation !!!!

physical interpretation

- coupling of gravity and electromagnetism in low dimension from gravity in high dimension
- The NS-NS action is $O(E, E)$ invariant.

prospect

- searching other symmetry
- considering an internal field dependence.

A matrix M is in $O(E, E)$ if it is a square $E \times E$ matrix and, for the matrix

$$\zeta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

which can be used as a metric for the group $O(E, E)$, as it has E negative eigenvalues and E positive eigenvalues in a basis rotated from the one with diagonal metric, the matrix M satisfies

$$M^T \zeta M = \zeta.$$